Minerals

Mineral occur in the Earth's crust and have 5 characteristics that make them minerals(criteria)

S_____

N_____ o_____

I_____

Fixed chemical f_____

Specific atomic arrangement

Ore=a rock containing a m_____. Minerals often contain specific elements. When elements need to be mined, geologists will look for the ore with the mineral they want.

Mineral	Element that it contains
Haematite	
Bauxite	
Cassiterite	
Pentlandite	

HARDNESS

The Mohs scale measures hardness of minerals. It has the idea that harder minerals can scratch softer minerals.

Diamond is the hardest mineral so it has a value of 10. It can scratch **all** other minerals.



Complete this table.

Hardness	Mineral
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Read the two worked examples on page 317. Then try these.

a. Unknown mineral X will scratch fluorite, but not feldspar. How hard is it?

b. Unknown mineral Z will scratch feldspar but not topaz. How hard is it?

c. Why is diamond given a Mohs hardness measure of 10?

d. Which minerals can talc scratch?

e. Which minerals can scratch talc?

f. Why is hard plastic not counted as a mineral?

g. Why is hard wood not counted as a mineral?

Minerals

A Z W M H I Y T K H P B J G E
L E N D G A J L O L Z R K Z I
U X O P N V R R L P O V S P J
M R E V I O Z D T A A G C N S

R E R O R W M S N Z R Z R L C
O N M Q R U U A N E J U A L I
F R Y Q U R Y G I J S R T N J
E R U T C U R T S D E S C A E
B R H E C P N S J N V S H L N
U I P L O Q Z I I J T M E D S
L A T S Y R C M G M O M B T R
R O J Z Q X O T T H E L P D E
Z X P V Y K D A S N F H G H B
B J W O X W Q L T D S O L I D
I N O R A N I C C A G E N E Q

CRUST
CRYSTAL
DIAMOND
ELEMENT
FORMULA
HARDNESS
INORANIC
MINERALS
MOHS
NATURALLY
OCCURRING
ORE
SCRATCH
SOLID
STRUCTURE
TALC
TOPAZ